Geometric Boundaries of the Frohmanian Symplectic Tether: Formal Tracking of the 2D Clock Test and Open Analytic Inputs for Fefferman Statement (B)

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/Clay-
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Abstract

Standalone formal boundary ledger (‘lean/Clay/*.lean‘, ‘latex/geometric_boundaries.tex‘, ‘latex/section10_gaps.tex‘, ‘latex/two_d_fact.tex‘, ‘COMPARATOR.md‘, ‘STATUS.md‘) analyzing the exact analytic boundary between algebraic symplectic-tether identities (Pi(u)=0, {F,E}=0) and unforced global smoothness on T^3 (Fefferman Statement B). Includes the 2D clock test showing that constant vortex direction (sin phi = 0) coexists with linear growth ||phi(t)||_infty = M_0^2 t on steady Euler equilibria.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Explicitly titled and structured as a boundary and gap ledger (‘STATUS.md: not a Clay resolution‘), isolating the open analytic obligations (unforced BKM stretching integral finiteness) without using ‘sorry‘.

- **Repository:** `https://github.com/BenFrohman/Clay-`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23653013`
- **Earliest Git Commit SHA:** `9e61806681bfce3f0edee28293d22a6a711f0780` (2026-10-07 06:16:45 AM CDT (UTC-5))
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2 Technical Overview & Repository Specification

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Private workspace for Fefferman statement (B): unforced 3D incompressible Navier–Stokes on $\{}mathbb{T}^3$.
This repository is not a completed Clay resolution.
- ‘lean/Clay/Gaps/TwoDClock.lean‘ — the 2D fact: constant direction, so Constantin–Fefferman holds, and the history integral still grows like $m^2 t$ wherever $M\{}ge m>0$. Steady Euler equality case $\{}|\{}phi(t)\{}|_\{}infty=M_0^2 t$.

- ‘lean/Clay/Gaps/MissingInputs.lean‘ — BKM integral, metriplectic/L2 gap, CZ/BMO gap, spike obstruction, holographic rejection. None closes Statement (B).

- ‘latex/two_d_fact.tex‘ — the same inputs in LaTeX.

- ‘COMPARATOR.md‘ — ‘FormalConjectures/Millennium/NavierStokes.lean‘. Not a dependency.

See ‘STATUS.md‘ for the Section 10 referee finding.

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/Clay-` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.js` and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
AUTHORS.md	267 B	d571bad8bbfeca08...
COMPARATOR.md	1198 B	a899cda8bd5c7160...
LICENSE	11355 B	9da72c2bee4f20ba...
NOTICE	317 B	80e5fa3638d62d2a...
README.md	903 B	d173785a4df45954...
STATUS.md	2491 B	252aae1e3a6a7243...
latex/geometric_boundaries.tex	5056 B	952c5705ca8851c5...
latex/section10_gaps.tex	2314 B	c43c9ac50665ba92...
latex/two_d_fact.tex	1490 B	b969949b2b80c90d...
lean/Clay/Depletion/GeometricAlignment.lean	490 B	4d7125ee4bc18711...
lean/Clay/Gaps/BKMIntegral.lean	511 B	19cea215acbc7903...
lean/Clay/Gaps/ConditionalRiccati.lean	693 B	b3c47df1a1cbeb43...
lean/Clay/Gaps/DirectionAlignment.lean	633 B	3395eae5672c0273...
lean/Clay/Gaps/HistoryClock.lean	662 B	b134ea98037b29fa...
lean/Clay/Gaps/MissingInputs.lean	795 B	010aed7022d8fcd7...
lean/Clay/Gaps/RejectedMeasureBound.lean	531 B	da60d8abc07fa028...
lean/Clay/Gaps/StretchingIntegral.lean	765 B	a1b919459c9bf774...
lean/Clay/Gaps/TwoDClock.lean	935 B	09bf6a419006c185...
lean/Clay/Gaps.lean	422 B	99171b435e416ff6...
lean/Clay/Tether.lean	1031 B	2e562a881a4d654a...
lean/Clay.lean	477 B	1d60fd986c2b3aba...
lean/lakefile.lean	577 B	3d3b19f370345872...
lean/lean-toolchain	25 B	b3f7d43e005c36e7...

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